ElderGuard - M3: Final Technical Report

ECE 310 - Wireless Communications, Monsoon Semester 2026
Milestone 3 - Minimum Viable Product (MVP) | Group 8

Camera-free Wi-Fi CSI People Counting Prototype

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Prototype scope
A two-node ESP32-S3-N16R8 link measures Wi-Fi channel state information (CSI). A laptop parses live receiver data and runs a trained classifier to estimate occupancy classes 0, 1, or 2. No camera is used.

Evidence note
This report distinguishes artifacts and console output supplied for the project from independently verified results. Hardware results are user-reported or taken from the shared run log; no physical test was conducted by the report author.

1. Executive summary

The M3 prototype implements the principal CSI people-counting path: Board A transmits Wi-Fi traffic, Board B receives packets and exports CSI over USB serial, and Laptop 1 performs data capture, preprocessing, model training, and live prediction. The demonstrated software path is operational in the supplied run log: CSI packets were received over COM15 and the model emitted 1- and 2-person estimates. The user also reports obtaining expected live outputs for 0, 1, and 2 people in room trials. These observations are promising demonstrations, not a statistically established accuracy claim. The saved training report remains provisional and does not contain independent-session validation.

2. Prototype description and architecture

Component	Implementation / role	Status
Board A / TX	ESP32-S3 access point ElderGuard-TX, fixed Wi-Fi channel, periodic UDP traffic. Laptop 2 powers/diagnoses TX.	Prototype firmware present; user run instructions specify COM4 for diagnostics.
Board B / RX	ESP32-S3 station joins TX AP, enables CSI reception and streams CSI rows over USB serial.	CSI receiver path was active in supplied live log.
Laptop 1 software	Python 3.11, serial collection, CSI parsing/features, Random Forest training, live inference and smoothing.	Dependencies installed; model trained; live inference launched.
Camera	Not used.	Out of scope.
Fall detection	Separate heuristic mode; occupancy classifier/data do not validate fall detection.	Not evaluated in this report.

Data flow: TX Wi-Fi packets -> RX CSI callback -> serial stream -> Python parser and temporal features -> classifier -> smoothed occupancy display. The ML model runs on Laptop 1, not on the ESP32 boards.

3. Requirements coverage

MVP requirement	Coverage	Evidence / gap
Functional prototype for intended capability	Partial / demonstrated	Live CSI stream and count estimates shown in console log; user reports expected 0/1/2 trials.
Document design, requirements, implementation status	Covered here	Architecture, status, limits, and next work described.
Test functionality, reliability, performance	Partial	Software self-test passed on synthetic fixtures per project history; real CSI console log shows reception and variable rates. No independent accuracy test.
Analyze limitations and failures	Covered	Single-session model, stale/slow stream warnings, empty CSVs, output variation, scope limited to 0-2.
Baseline comparison and revised SOTA positioning	Analytical only	Qualitative positioning below; no controlled head-to-head benchmark was supplied.

4. Experimental setup and dataset

Hardware configuration specified in project materials: two ESP32-S3-N16R8 boards. Laptop 1 (DELL) is RX/COM15 at 921600 baud for Python capture and inference. Laptop 2 (Admin) is TX/COM4 for diagnostics and

power. The transmitter AP is configured as ElderGuard-TX on channel 1. Board placement should remain fixed between collection and evaluation.

Capture	Class	Frames	Usable status per saved training run
roomA_0p_t01.csv	0 people	4,578	Loaded
roomA_1p_t01.csv	1 person	5,808	Loaded
roomA_2p_t01.csv	2 people	5,563	Loaded
diagnostic.csv	Unlabeled	present	Skipped for supervised training
roomA_2p_t01_fs60.csv; roomA_2p_t04_fs60.csv	Unlabeled/empty by parser	0 parsed rows	Skipped

Folder inspection also found non-empty roomA_0p_t02.csv and roomA_1p_t02.csv. The saved model report predates or excludes them: it lists only t01 sessions. The current status of the t02 recordings, their ground-truth protocol, and a retrain using them must be confirmed before treating them as training or validation data. No t02 two-person recording was listed as non-empty.

5. Signal processing and model

The documented parser reads signed-byte CSI vectors, uses the first 64-complex-sample LLTF section consistently, rejects malformed odd-length vectors, and estimates sampling rate from receiver timestamps. A one-second temporal window with 50% overlap yields 499 windows for the original three captures. Twelve described features include CSI amplitude/phase variation, subcarrier correlation, Doppler-related measures, RSSI statistics, and packet delivery ratio. A Random Forest was selected as the predeclared baseline and saved as Results/model_count.joblib.

The saved training_report.json says LeaveOneRecordingOut validation is unavailable, with one recording per class; it marks the artifact provisional and its model scores uncalibrated. It reports no held-out accuracy. Window counts are not independent experiments because adjacent windows from the same recording are correlated.

6. Test results

Test / observation	Result	Interpretation
Python environment and dependency install	Python 3.11.9; required packages already installed (reported terminal output).	Setup successful.
Dataset validation	Six CSV files found; expected files present (reported terminal output).	Validation utility ran; inspect generated report for per-file findings.
Training	499 feature windows; Random Forest saved; provisional status.	Training pipeline ran, but generalization is not measured.
Live CSI receiver	COM15 output showed packet rates typically about 27-63/s and RSSI around -42 to -52 dBm; intermittent low-rate/stale warnings.	CSI path active but stream stability varied.
Live occupancy inference	Supplied log alternated estimates between 1 and 2; user separately reports exact 0/1/2 physical trials.	Demonstrates inference capability; repeat trials and independent sessions needed for reliability claim.

No accuracy percentage, fall-detection performance, or 100% correctness is claimed. The project has not been independently tested on additional rooms, layouts, distances, or counts above two.

7. Baseline comparisons and revised SOTA positioning

The project baseline is a laptop-side Random Forest using handcrafted temporal CSI features, compared with a trivial majority-class prediction only conceptually; no numerical baseline benchmark has been run. The appropriate next comparison is grouped, session-held-out evaluation of a majority-class baseline, Random Forest, and one or two lightweight alternatives (for example SVM and k-nearest neighbors), using identical session folds and reporting macro-F1, balanced accuracy, confusion matrix, and per-class recall. Do not randomly split overlapping windows across train/test.

Relative to camera-based counting, this system is camera-free and can operate without image capture, but it estimates only coarse occupancy classes from a device-specific RF link. Relative to established Wi-Fi CSI research systems, the current MVP uses only two commodity ESP32-S3 boards, one room/link, a small dataset, and a laptop model. It lacks multi-room evaluation, a large labeled corpus, calibrated uncertainty, demographic/position robustness analysis, and a controlled published-method benchmark. Therefore it should be positioned as an early low-cost prototype, not state-of-the-art or deployment-ready.

8. Limitations, failed cases, and risks

1) Very limited independent recordings; saved model is provisional. 2) In the supplied live log predictions change between one and two people. 3) CSI packet throughput varies and stale-window warnings occur. 4) Two zero-byte CSVs are unusable. 5) Exact receive-side CSV schema must remain consistent with the Python parser; an RX sketch pasted during development appeared to have a header/row field-count mismatch and should be compile- and capture-checked before flashing. 6) Results are room-, geometry-, channel-, hardware-, and environment-dependent. 7) Only 0/1/2 classes are supported. 8) Fall detection is separate and unvalidated.

9. Further work and test plan

Before final performance claims: (a) audit each capture for truth label, frame count, timestamps, RSSI, and gaps; (b) collect at least 5-10 distinct sessions per count across varied positions and still/moving conditions while maintaining the chosen deployment geometry; (c) reserve complete sessions for testing; (d) compare grouped cross-validation models and report confusion matrix and per-class metrics; (e) conduct repeated live trials with fixed trial duration and a tally of predictions/errors; (f) report packet rate and stale intervals; (g) repeat in a second room/layout if generalization is claimed. A practical minimum for a stronger initial evaluation is 5 sessions per class, but this is a target, not a guarantee of accuracy.

For each live trial record: timestamp, actual count, predicted count, fraction of displayed windows correct, packet rate, RSSI, board positions, and environmental changes. Keep raw CSVs immutable and use unique filenames.

10. Reproducibility commands

From the project root on Laptop 1:

```
py -3.11 -m pip install -r requirements.txt
py -3.11 tools\validate_dataset.py
py -3.11 tools\train_model.py --data-dir Data\classroom --out-dir Results\m3_retrain
py -3.11 tools\live_demo.py --port COM15 --baud 921600 --mode count --model
Results\m3_retrain\model_count.joblib
```

Close Arduino Serial Monitor before Python opens COM15. Keep TX active on Laptop 2. Use the actual COM number shown in Device Manager if Windows changes it.

11. Conclusion

The team has assembled a camera-free ESP32-S3 CSI people-counting prototype with dataset tooling, model training, and a live laptop inference path. The shared logs demonstrate that the pipeline runs and receives CSI. The evidence is not yet sufficient to establish dependable counting accuracy: the saved model report is provisional and live logs include 1/2 fluctuations and intermittent stream gaps. The next milestone is an independent-session

evaluation with more truthful labeled recordings and transparent error reporting.

Appendix: submission contents

Recommended GitHub M3 folder: M3_Group8/README.md; M3_Group8/Report/M3_Group8_Report.pdf; project source folders src/, tools/, firmware/; requirements.txt; selected configuration and documentation; and small, ethically shareable dataset samples or a dataset acquisition note. Preserve code needed to reproduce the work. Do not commit generated virtual environments, Python caches, serial logs containing private information, secrets, or large raw recordings unless permitted and useful; include dataset provenance, labels, and a download/access note if raw data is excluded. Include a clear README with hardware, wiring/ports, setup, training, live-run commands, dataset summary, and limitations. Each member completes the individual Google Form separately. Maintain meaningful incremental Git commits; never fabricate or backdate history.

Source basis: project README, saved Results/training_report.json, Data/classroom file inventory, and user-shared PowerShell run output. User-reported physical outcomes are identified as such. No external claims or hardware tests were independently verified for this report.